Github – Maternal Health Data Project Group 3

Team member

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Dataset Information

Number of samples: 1,014

- Age: Age in years when a woman is pregnant.
- SystolicBP: Upper value of Blood Pressure in mmHg, another significant attribute during pregnancy.
- DiastolicBP: Lower value of Blood Pressure in mmHg, another significant attribute during pregnancy.
- BS: Blood glucose levels is in terms of a molar concentration, mmol/L.
- HeartRate: A normal resting heart rate in beats per minute.
- Risk Level: Target variable, devided into three categories: Low risk, Mid risk, High risk. Predicted Risk Intensity Level during pregnancy considering the previous attribute.

About the Project

This is a project for the subject Programming for Data Analytics at Rennes School of Business. This dataset focuses on maternal health risk — the likelihood of a pregnant woman experiencing complications or negative outcomes related to pregnancy or childbirth. It was collected to support research in public health and predictive analytics.

This project uses machine learning to predict the risk level of maternal health during pregnancy (low, medium, or high) based on vital health indicators. The goal is to help identify women who may be at risk so that doctors and health workers can take preventive actions early.

About the Dataset

The datasets have been obtained through Kaggle platform and selected to provide relevants insights regarding Maternal Health Risk Data that have been collected from different hospitals, community clinics, maternal health cares through a risk monitoring system. The data collected includes insights on age, systolic blood pressure, diastolic blood pressure, blood sugar, heart rate and risk level. The risk level is presented in three categories: high risk, medium risk, low risk. The dataset has a total of 1014 entries in the dataset, with a total of 37 outliers that were removed before further analysis.

Project Steps

We started by importing the dataset using Python and examining its structure with pandas to understand the number of rows, columns, and data types. We performed data cleaning by checking for missing values, removing duplicate rows, and confirming that all numeric and categorical columns were correctly formatted. Exploratory Data Analysis (EDA) was conducted using histograms, boxplots, and correlation heatmaps to observe the distribution of features and how they relate to the target variable, RiskLevel. Outliers were identified using z-scores and removed from the dataset to prevent extreme values from skewing the model. The categorical target variable, RiskLevel, was encoded into numeric labels to allow machine learning models to process it. We standardized the numeric features using StandardScaler so that all features were on a comparable scale, which improves model performance.

The data was split into training and testing sets using a 70/30 ratio, ensuring that the model could be evaluated on unseen data. A Random Forest Classifier was trained to predict maternal risk levels, leveraging its ability to handle multiple features and classify into three categories. The model was evaluated using metrics such as accuracy, precision, recall, and F1-score to measure how well it predicted each risk category. We analyzed feature importance to determine which health indicators had the greatest influence on maternal risk, and we suggested possible threshold values for higher-risk patients.

Main Results

The EDA indicates high blood pressure and blood sugar are strong indicators of higher maternal risk. Risk increases slightly with age. The classes (Low, Mid, High) are fairly well balanced.

37 outlier rows were removed using the Z-score method (values more than 3 standard deviations from the mean). This helps prevent extreme values from distorting the model.

The Random Forest model achieved an overall accuracy of approximately 85%, indicating that it can reliably classify low, medium, and high maternal risk.

The precision, recall, and F1-score were balanced across the three classes, showing that the model performs well for all risk levels without heavily favoring one class over another.

Analysis of feature importance revealed that blood sugar levels, systolic and diastolic blood pressure, and age were the most influential factors in predicting maternal risk.

Based on the high-risk group, we identified possible thresholds for each health indicator. For example, patients over 35 years of age, with systolic blood pressure above 120 mmHg, diastolic blood pressure above 90 mmHg, blood sugar above 11 mmol/L, body temperature above 98°C, or heart rate above 78 bpm are more likely to be in the higher-risk category.

These results suggest that monitoring these key indicators can help identify high-risk patients early, allowing healthcare providers to take preventive measures and improve maternal health outcomes.

Prescriptive Insights

If blood sugar or blood pressure values exceed the identified thresholds, patients should be monitored more closely. Regular check-ups and early medical interventions could help prevent complications. Hospitals and clinics could use such models to support preventive care programs. The model can be further improved with larger and more diverse datasets.

Technologies & Tools

- Python (Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn)
- Jupyter Notebook
- Visual Code Studio